The benchmark is the binding constraint

A registered comparison of seven portfolio constructions, four defences against fooling yourself, and a reproduction claim that was wrong twice

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Status: EXPLORATORY AND NON-CONFIRMATORY. The registered test was executed without the blindness its registration required, so this run may not be cited for or against either hypothesis: the hypotheses are **untested, not falsified**. The verdict the frozen rule returns is quoted below for completeness; it is not a test of anything. The description this work asks to be held to is *pre-registered, mechanically evaluated, non-confirmatory*. Net **excess** Sharpe only, at stated costs, with intervals; no absolute return claim; nothing here is an investable return.

Abstract. This paper compares 7 constructions for spreading a portfolio across 18 US equity, bond, commodity and real-estate ETFs over 168 out-of-sample months. Two of them are treatments, and each differs from a sample-covariance, sample-mean mean–variance baseline (`B3`) in exactly one component: `M1` replaces the sample covariance with a Ledoit–Wolf shrinkage estimate; `M2` replaces the sample mean with a Black–Litterman posterior. The decision rule was fixed before the run and is a conjunction — a net excess Sharpe advantage of at least 0.20 over the best baseline, a probabilistic Sharpe ratio of at least 0.95 against that same baseline, and six robustness gates — not a ranking. Both treatments fail. The larger advantage, 0.064 against a bar of 0.20, is a point estimate whose interval overlaps the baseline's, and it belongs to the treatment that changes the mean rather than the covariance. A post-hoc cap sweep does not rescue the shrinkage treatment; the mean treatment's advantage grows as the cap loosens, but remains below the bar. The same two strategies clear a deflated-Sharpe test against zero at $p = 0.0017$ and 0.0032, which places the binding constraint in the declared benchmark rather than in the multiple-testing correction. Because validating the reporting layer required running the frozen configuration first, no blind test exists: the run is exploratory and the hypotheses are untested, not falsified. The paper's second subject is the instrument — what a reproduction check can prove, and what it cannot.

1 The question

Robust portfolio construction is usually sold as a cure for estimation error. The argument is old and sound [1]: a mean–variance optimiser treats estimated means and covariances as if they were known, so it loads on the assets whose estimates are most optimistic, and the portfolio it produces is often worse out of sample than equal weight [2]. The proposed cures — shrink the covariance [3], put a prior on the mean [4] — are well motivated. The evidence offered for them is usually a backtest in which the cured portfolio has a higher Sharpe ratio.

This paper takes that last step seriously. A higher number is not evidence unless four things hold: the comparison changes one thing; the rule that would reject the improvement was fixed before the result was seen; the improvement is large enough to matter; and the numbers can be checked by someone who does not trust the author. The first three are design choices. The fourth is an engineering one, and it is the one this project spent most of its effort on.

The result here is a null. Two treatments that differ from a mean–variance baseline in exactly one component each — a shrinkage covariance and a Black–Litterman mean — produce no net excess Sharpe advantage large enough to clear a bar fixed in advance. The paired test is powered to detect

an effect at that bar; the registered level gate is not. Both fail the registered rule. Neither is rescued by a cost sweep or by the fold structure, and the intervals do not separate them from the baseline.

Four things are contributed, and only the first is a result:

1. a single-factor registered comparison, in which each treatment differs from the baseline in one component, so a difference is attributable to that component rather than to a bundle;
2. evidence that under a single registered comparison the deflation machinery is inert, and the **declared benchmark** is what decides the test — which makes fixing the comparator a statistical requirement rather than bookkeeping;
3. a reproduction protocol that splits the claim in two: decisions compared exactly, measurements compared within a bound, with the observed difference printed on every run;
4. a reported null, with the rule that produced it, the reading it depends on, and the circumstances under which it would have said otherwise.

What this is not: a claim of tradable alpha, a confirmatory test, or a survey. The sample is one universe, one period, one cost model, and the hypotheses remain untested.

2 Design

2.1 Universe and sample

The universe is 18 US-listed ETFs spanning equity, fixed income, commodities and real estate. Membership is reconstructed from a date-stamped candidate list in which each fund carries the date from which it is eligible, so no instrument enters before it existed. This is point-in-time **eligibility**, not point-in-time **data**: the prices are what a 2026 vendor holds for 2012 today, and price history is revised. The distinction is not repaired here and it is not small.

Rebalancing is monthly, on the last trading day. The estimation window is 60 months, rolling and re-estimated every month; test windows are twelve months and do not overlap. Portfolios are fully invested and long only, with a uniform 30% cap on any position, and the mean–variance constructions share a risk aversion of 10. Costs are charged one way at a headline 10 basis points, and the run reports every metric at 0, 5, 10, 20, 50 basis points so that the headline is a choice among reported levels rather than a number selected after the fact.

The sample begins with the first eligible month, in mid-2012, and ends in September 2026: 168 monthly rebalances in 14 non-overlapping twelve-month folds, and 1176 optimiser solves across the 7 constructions.

2.2 Seven constructions

Five are baselines and two are treatments. The treatments are single-factor by construction: `M1` is `B3` with the sample covariance replaced by a Ledoit–Wolf shrinkage estimate; `M2` is `B3` with the sample mean replaced by a Black–Litterman posterior. Universe, window, optimiser, constraints, costs and rebalancing are shared. A difference between a treatment and `B3` is therefore attributable to the one component that changed — and that minimal-difference design is also what buys the statistical power, because the paired comparison of two almost-identical return series is far more sensitive than the comparison of either to zero.

ID	Construction	Differs from `B3` in	Role
B1	Equal weight	no optimiser	Primary benchmark
B2	Risk parity	equal risk contribution	Practitioner default
B3	Sample MVO	—	Critique target
B4	Minimum variance	no mean	Robust alternative
B5	Turnover-penalised MVO	an L1 penalty	Robust baseline
M1	Shrinkage-covariance MVO	the covariance	Treatment, H2.1
M2	Black–Litterman MVO	the mean	Treatment, H2.2

Table 1: The seven constructions. `M1` and `M2` each change one component of `B3`, so a difference between a treatment and the baseline is attributable to that component; nothing else differs, including the constraints and the cost model. `B3` was the best baseline in this run.

2.3 The decision rule

The registered rule is a conjunction of eight gates, applied to the best baseline rather than to zero:

1. a probabilistic Sharpe ratio against the best baseline of at least 0.95 — the probability that the candidate's Sharpe exceeds the baseline's;
2. a deflated-Sharpe p-value of at most 0.05;
3. a net excess Sharpe advantage over the best baseline of at least 0.20;
4. a turnover ratio against the baseline of at most 2.00;
5. a drawdown slack of at least −0.050;
6. a win at 3 of the 5 cost levels;
7. a win in at least 9 of the 14 folds;
8. no single fold contributing more than 40% of the advantage.

Every threshold is read from the published verdict table rather than restated by hand, and Table 3 prints the rule gate by gate for both candidates. The rule is a representation of the decision, not the decision itself: it compresses a return path, a drawdown path and a set of folds into eight gates. Whether that representation answers the question the design was built for is a separate question, and it is the one the correction turns on.

Significance alone admits improvements too small to act on; materiality alone admits noise. Requiring both, plus robustness across cost levels and folds, is what makes a null informative: a candidate that fails has failed at the bar the design was built to detect, not at an arbitrary one. Two caveats belong with the rule rather than in a footnote. The fold gate is weak: if folds were coin flips, 21.2% of null experiments would pass it on their own, and the folds share overlapping training windows, so the true rate is higher. And the materiality gate is a decision about the world, not a statement about noise: it says an improvement is too small to act on, which is why the bar is stated in the units the decision is made in.

2.4 What was fixed before the run

The registration fixed the hypotheses — H2.1, that a shrinkage covariance beats the sample covariance on net Sharpe, and H2.2, that a Black–Litterman posterior beats the historical mean — together with the metric, the thresholds, the headline cost level, the fold structure and the conjunction above. It also fixed the comparator **mechanically**: the rule names the best baseline, where "best" is defined as the highest net Sharpe among the five baselines. The identity of that baseline is an output of the run, but the definition that selects it is not, so the comparator cannot be chosen by hand after the answer is known. Section 3.2 argues that this distinction carries most of the weight; Section 6 records what it does not settle.

What the registration did not fix, because it could not, is the order in which the reporting layer and the test would run. Section 4 is about that.

3 Results

3.1 The null

Table 2 and Figure 1 report the same estimates. `B3`, the plain sample-covariance, sample-mean optimiser that the project set out to critique, is the best baseline at 0.7896. `M2` has the highest point estimate of any construction, 0.8538, an advantage of 0.064. That is below the 0.20 bar. `M1` lands on `B3`: its advantage is −0.004, which is to say none. The point estimate is a summary of one historical path, not a forecast; the interval is the uncertainty around that summary, not a range of possible futures.

Method	Net excess Sharpe	90% interval	Turnover	Max drawdown
M2	0.8538	[0.426, 1.369]	2.14	−17.6%
B3	0.7896	[0.331, 1.318]	2.12	−18.4%
M1	0.7860	[0.327, 1.298]	1.92	−18.0%
B5	0.7843	[0.316, 1.316]	0.73	−19.8%
B1	0.5482	[0.153, 1.002]	0.39	−18.1%
B2	0.4519	[0.017, 0.974]	0.34	−13.4%
B4	0.1820	[−0.253, 0.751]	0.43	−10.9%

Table 2: Net excess Sharpe at the 10 bps one-way headline cost, with a 90% stationary-bootstrap interval (2000 replications, mean block 6 months), annualised turnover, and maximum drawdown of the net excess series. Candidates are bold. Exploratory and non-confirmatory: no confirmatory test exists, and these numbers may not be cited for or against either hypothesis.

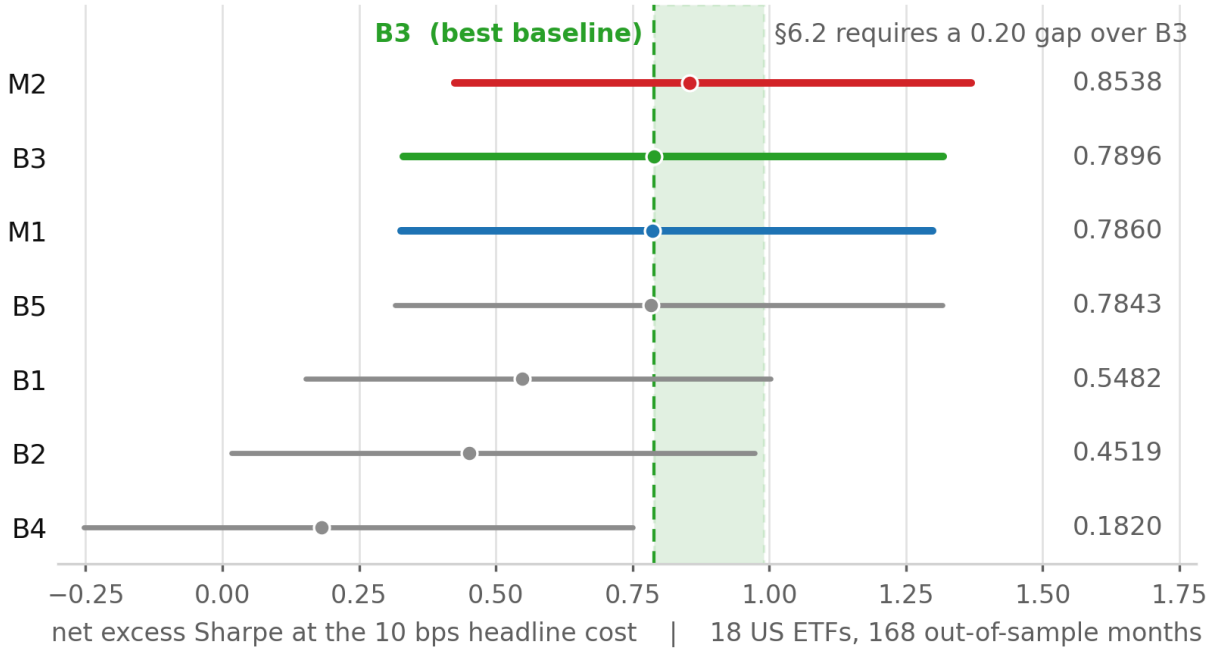


Figure 1: The same seven estimates as intervals. The shaded band is the distance the materiality gate requires — 0.20 of net excess Sharpe to the right of B3 — so a candidate clears it only if its dot sits to the right of the band. Intervals overlap for every pair of constructions. Exploratory and non-confirmatory.

The intervals are the result, and the ordering is not. Every construction's 90% interval overlaps the interval for `B3`, most of them by a wide margin, so the table's ordering by point estimate is a ranking of estimates rather than a separation of strategies. The two treatments carry as much turnover as the baseline or more.

Both treatments fail the rule. Table 3 shows why, and the failures are not marginal: `M2` passes five of the eight gates and fails three, `M1` passes three and fails five. The gate that decides both is the

probabilistic Sharpe ratio, where they score 0.495 and 0.587 against a bar of 0.95. The concentration gate also fails for both: even where an advantage exists, it lives in a single fold — 50.5% for `M1` and 51.3% for `M2`, against a 40% limit.

`M2` is worth a second look, because it is the case where a careless report would have found a win. It leads the ranking, wins 5 of the 5 cost levels and 10 of the 14 folds, and clears the deflated-Sharpe gate. It still fails on the two gates that ask whether the advantage is **worth having**: the gap and the probability that the gap is real. That is the conjunction working as designed — the robustness gates are satisfied and the materiality gate is not, so the answer is no.

One reading is load-bearing and is disclosed rather than buried. The protocol requires the advantage to hold across folds but does not pin "advantage" to a metric. The reading used here — fold wins measured on net Sharpe, the concentration clause measured on the per-fold mean-return difference — is the reading that decides the verdict for `M2`, because the concentration gate fails at 51.3%. Whether a different reading would change the verdict is a computation that has not been run, and this paper does not claim it would. The verdict is therefore the output of a stated reading, not of the arithmetic alone.

Gate	M1	M2
PSR against the best baseline, at least 0.95	**fail**	**fail**
Deflated-Sharpe p-value, at most 0.05	**pass**	**pass**
Net Sharpe gap over the best baseline, at least 0.20	**fail**	**fail**
Turnover ratio, at most 2.00	**pass**	**pass**
Drawdown slack, at least −0.050	**pass**	**pass**
Cost levels won, at least 3 of 5	**fail**	**pass**
Folds won, at least 9 of 14	**fail**	**pass**
Largest fold share, at most 40%	**fail**	**fail**
Passed	3 of 8	5 of 8
Verdict	**FAILURE**	**FAILURE**

Table 3: The registered conjunction. A candidate passes only if all eight gates pass. `M2` clears the statistical gate, the robustness gates and the concentration clause's siblings, and fails the two gates that ask whether the advantage is large enough to matter.

3.2 The benchmark, not the correction, is the binding constraint

Both treatments clear the deflated-Sharpe test against zero: p = 0.0032 and 0.0017. Both fail the probabilistic Sharpe test against `B3`: 0.495 and 0.587. Same return series, opposite conclusions, and the only difference is what each statistic is measured against.

That is not a paradox. With exactly one registered comparison the family size is one, so the selection deflation is inert: the deflated Sharpe ratio and the probabilistic Sharpe ratio differ only in their reference point, zero in one case and `B3` in the other. The machinery the project specified to correct for multiple testing never fires, because with one test there is nothing to correct. What remains binding is the benchmark — the one input to the test that the analyst chooses, and the input for which, after the fact, there is always a flattering value.

Both p-values answer "is this strategy's Sharpe greater than zero", not "is it better than the baseline". A reader who takes p = 0.0017 as evidence of improvement has answered a question nobody asked. This is the sentence the project exists to make difficult to skip.

3.3 The cost sweep, and the turnover inversion

Table 4 reports the same metric at each of the 5 cost levels. Across them the advantage of `M2` over `B3` moves between 0.064 and 0.065; the advantage of `M1` between −0.007 and 0.012. The bar is 0.20 at every level. Sweeping costs does not make a null true, but a conclusion that reversed between two adjacent cost levels would not be worth an interval, and this one does not reverse.

Method	0 bps	5 bps	10 bps	20 bps	50 bps
M2	0.8791	0.8665	0.8538	0.8284	0.7519
B3	0.8151	0.8024	0.7896	0.7640	0.6870
M1	0.8077	0.7968	0.7860	0.7642	0.6986
B5	0.7926	0.7885	0.7843	0.7760	0.7509
B1	0.5527	0.5504	0.5482	0.5436	0.5299
B2	0.4592	0.4556	0.4519	0.4447	0.4228
B4	0.1953	0.1887	0.1820	0.1687	0.1285

Table 4: Net excess Sharpe at each reported one-way cost level. The headline level is 10 bps. Exploratory and non-confirmatory.

The usual story for robust construction is that it wins by trading less. Here it inverts. `B3` carries 2.12 annualised turnover and still leads on net excess Sharpe at the headline cost, while `B5`, the turnover-penalised baseline, cuts turnover to 0.73 and lands below `B3`. The two treatments carry as much turnover as the baseline or more. In this sample, turnover penalisation bought a real reduction in trading and did not buy an improvement in net Sharpe, while the highest-turnover construction was the best performer. That is a legitimate outcome, and it is the kind of thing a single headline number would have hidden.

3.4 A diagnostic hypothesis, labelled as one

Eighteen assets are not eighteen bets. On the frozen window the effective number of independent directions is 3.14 by participation ratio and 5.03 by entropy — far below 18. The near-singularity is structural rather than a short-sample artefact: for an independent panel of this shape the Marchenko–Pastur band would be [0.452, 1.762], and 12 of the 18 eigenvalues fall below the lower edge, the smallest at 0.0053, about 85 times below it [8].

If that is the right reading, shrinkage has little independent noise to bite on, and the uniform 30% cap is doing more regularisation than the estimator is. That would explain why `M1` lands on `B3`, and why `M2` — the only treatment that changes the mean rather than the covariance — is the only one that moves the point estimate at all.

It is a hypothesis, and it is labelled as one. The cap sweep that tests it, across four values of the cap, was registered and has since been run (2026-11-04): it does not support the shrinkage version. `M1′`s difference from `B3` stays at or below zero as the cap loosens. The cap does bind the Black–Litterman treatment, whose advantage grows with the cap but remains below the materiality bar. One arm — the turnover-penalised method at the two looser caps — could not be certified at the frozen tolerance and is reported as incomplete rather than tolerated. The test is reported in the repository note, not as a result of this paper.

4 The defence that failed

Four defences were built before the result was read. Freeze the metric, thresholds and decision rule before looking, so the comparator cannot be chosen after the answer is known. Prove the absence of leakage mechanically, with a test suite rather than a code review — including one test recorded as vacuous rather than passed. Certify the numerics: every rebalance's KKT residual checked against a frozen tolerance, 1176 solves with zero rejections and a worst residual of 1.49e-11 against a 1e-7 bar. And require significance, materiality and robustness together rather than any one of them.

Three of the four held. The first failed, and it failed for the most ordinary of reasons: the reporting layer had to be validated before any result could be read, and the only configuration that could validate it was the frozen one — which was the registered one.

State that precisely, because the difference between an accident and a design failure matters. The registration required that the confirmatory test be executed blind: the analyst must not know the answer when the test runs. Validating the generator required re-running the harness on the frozen configuration and reading its table. The frozen configuration is the confirmatory configuration. So the answer was on screen, in a file being diffed, before the test ran. Nothing was edited; no threshold moved; the rules that ran were the ones written down. The blindness was gone.

The consequence is not cosmetic. A test that was not blind is not a confirmatory test, and blindness cannot be restored to a sample whose answer has been seen. The run is therefore exploratory and non-confirmatory, and the hypotheses are **untested, not falsified** — a distinction the project's claims policy fixes in advance, precisely so that a result like this one cannot later be relabelled into support. The registered rule's verdict stands as an output of the rule, and is not evidence about H2.1 or H2.2 in either direction.

What survives is the design, the numbers and the disclosure. What does not survive is any claim that either hypothesis has been tested. A confirmatory statement would need a sample that does not yet exist, registered before it does.

5 Reproduction: what a check can prove

The numbers in this paper are checkable because the repository ships the frozen archive and a verifier that recomputes both tables from it. That instrument makes a stronger claim than "the code runs": it compares. The claim it made first was too strong, and the check itself falsified it twice.

5.1 Two failures

The original claim was exact equality — the same bytes and the same code produce the same table, on any machine. The first run of the comparison found one column, an autocorrelation, differing by 4.163e-17: three units in the last place. The cause was not the arithmetic but the kernel. The correlation went through a BLAS routine whose implementation is selected at run time from the CPU's feature set, so the same expression rounded differently on a different CPU. The fix was to compute that column with exactly-rounded summation, so that every rounding step is defined by IEEE-754 rather than chosen by a kernel.

The second run failed harder: thirteen of the eighteen columns moved, the worst by 4.698e-12. That one is not a defect and cannot be fixed the same way. The covariance is a matrix product evaluated by a runtime-chosen kernel, and a quadratic program is solved only to its tolerance — a perturbation of one part in 10^{16} in the input reappears in the weights at about 10^{-12}. Asking a solver's output to be bit-for-bit equal across machines asks for a property the artefact never had.

5.2 The split

The response was not to widen the tolerance until the check passed. It was to notice that the claim had been about two different kinds of object. Decisions are discrete: which methods ran, how many rebalances and folds, the headline cost level, the ranking, the selected baseline, the eight gates and the verdict. Measurements are continuous: the metrics and the statistics the verdict was computed from. The first set is compared exactly, and any difference fails the build. The second is compared within 1e-9, and the difference actually observed is printed on every run rather than the bound being quoted. The bound is roughly two hundred times the worst cross-machine difference measured, and five orders of magnitude below the precision at which these numbers are reported.

That split is this section's methodological claim, and it generalises beyond this project: a reproduction claim should be stated separately for the objects that cannot wobble and the objects that can. "It reproduces" is not one claim.

5.3 A check that can be seen refusing

A check that has never failed is untested. This one has failed twice on real data, and the repository also ships a demonstration that tampers with in-memory copies of the published tables to show the refusal working: a measurement moved by 0.01 is refused and the column named; a verdict flipped from `FAILURE` to `SUCCESS`, with every statistic untouched, is refused with `VERDICT differs`; and a difference of 10^{-12}, the scale a different machine produces, is absorbed, because a check that fails on platform noise is useless. The first two are the behaviour that matters and the third is what makes the check usable.

One limit remains, and it is worth stating rather than leaving to be found. No real run has ever disagreed with the published verdict, so the verdict-refusal path is demonstrated synthetically and not on a genuine disagreement.

6 Limitations and what would change the verdict

The sample. One universe of 18 funds, one period of 168 out-of-sample months in 14 folds, one asset-class family. The design can separate a net excess Sharpe difference of roughly 0.13 to 0.21 between these highly correlated strategies, and it is silent about smaller effects. The null is therefore "no effect at or above the bar", not "no effect". A true improvement of half the bar would not have been detected here and must not be reported as though it had been. The backtest is one path through one period, not a forecast of the next one.

The comparator. The rule compares against the best baseline, identified mechanically after the run. The definition is fixed and the identity is not, which is weaker than naming the comparator in the registration. In this sample `B3` was the best baseline by a margin of 0.0053 over the next one. That is a narrow margin: the comparator choice is a fragility of this run, not a property of the design, and `M1′`s sign already depends on it.

The cost model. A flat one-way cost with no market impact, capacity, taxes or regime dependence. Costs cannot reach the optimiser: each rebalance is solved once and every cost level is derived from that single turnover series, so a cost level cannot influence the weights it prices. A capacity-constrained version would likely penalise the two high-turnover treatments, which would make the null stronger rather than weaker — but that is an argument, not a measurement.

The reading. One clause of the rule is not pinned to a metric, and the reading used here decides one candidate's verdict. A different reading is a different computation; it has not been run, and the paper does not claim it would change the result.

The status. No confirmatory test exists, so nothing here is evidence for or against H2.1 or H2.2. This is the binding limitation: the other four qualify a null, and this one says the null is not a test.

What would change the verdict: a confirmatory replication on a sample that did not exist when it was registered, with the same rule and the same bar; a cap sweep that isolates how much of the regularisation comes from the constraint rather than the estimator; a wider universe; and a cost model with impact. An effect at or above the bar, with an interval that excludes the baseline, would also do it — and would be the outcome the design was built to detect.

7 Reproducing this paper

Every number in Table 2, Table 4 and Table 3, and in Figure 1, is read from `docs/published-run/` by `scripts/build_paper.py`. The paper's source contains no typed measurement, and a test checks that by searching it for the values in the tables. To check the tables against the release archive:

```
python scripts/verify_release.py \
    --archive <the release asset> \
    --run docs/published-run \
    --expect-sha256 <the digest quoted in the release notes>
```

The verifier exits zero on `EXACT` or `WITHIN #sourced.tolerance`, and non-zero on any difference in a decision, including the verdict. To regenerate this document:

```
python scripts/build_paper.py
```

If the verifier reports anything other than those two outcomes, that is the interesting result.

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